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Pathway: Becalos x TechnoReady.

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Title of the Challenge: Server and Database Commands

SerApi

SerpApi is a real-time API to access Google search results. We handle proxies, solve captchas, and parse all rich structured data for you. Is a service that provides an Application Programming Interface (API) to retrieve structured data, such as JSON, from search engine results pages (SERPs) across various search engines, including Google and Bing.

SERP (Search Engine Results Pages) APIs access search engine data programmatically. They mimic user searches, extract results (URLs, titles, snippets), and deliver them in structured formats (JSON, HTML). They allow real-time insights, competitive analysis, and SEO research through various search parameters and locations.

Google Scholar API.

Google Scholar API allows to scrape SERP results from a Google Scholar search query. The API is accessed through the following endpoint: `/search?engine=google_scholar`. A user may query the following:

`https://serpapi.com/search?engine=google_scholar`

Utilizing a GET request. Head to the playground for a live and interactive demo.

Endpoints.

SerpApi provides several Google Scholar-related endpoints (via the `/search` endpoint with different engine parameters) for different kinds of queries:

Endpoint	Description.
<code>/search?engine=google_scholar</code>	General Google Scholar organic search (articles, journals) SerpApi
<code>/search?engine=google_scholar_cite</code>	Retrieve "Cite" / citation references (i.e. documents citing a given work) SerpApi
<code>/search?engine=google_scholar_author</code>	Query author / profile pages and their publications SerpApi+1
<code>/search?engine=google_scholar_author_citation</code>	Fetch the detailed citation info for a particular author's publication (with <code>view_op = view_citation</code>) SerpApi

Authentication Methods.

Access to the API requires an **API Key**, provided upon registration at SerpApi.

- The key is appended to requests using the parameter:

`api_key=YOUR_API_KEY`

- Every request must include this parameter.
- The key identifies the user, enforces quota limits, and prevents unauthorized access.

This ensures secure and accountable usage across all client applications.

Query Parameters

The API accepts a wide range of parameters to refine searches. Common options include:

- *q* : Search query (e.g., `q=deep+learning`).
- *hl* : Language of the results interface (default: `en`).
- *num* : Number of results per page (max: 20).
- *start* : Pagination offset (e.g., `start=20` for results starting from the 21st).
- *as_ylo* : Restrict results to publications after a specific year.
- *as_yhi* : Restrict results to publications before a specific year.
- *cites=<id>* : Retrieve articles citing a specific publication.
- *cluster=<id>* : Access versions or related works of a publication.

Response Formats

Responses are structured in **JSON**, making them easy to parse across programming environments. A typical response includes:

```
{
  "search_metadata": { "status": "Success" },
  "search_parameters": { "q": "deep learning" },
  "organic_results": [
    {
      "title": "Deep Learning in Medical Imaging",
      "link": "https://...",
      "snippet": "Overview of applications...",
      "publication_info": {
```

```

        "authors": "...",
        "year": 2020
    },
    "inline_links": {
        "cited_by": { "total": 532 }
    }
}
]
}

```

- *search_metadata*: Information about the request.
- *search_parameters*: Parameters used in the query.
- *organic_results*: Core search results, including titles, authors, publication year, links, and citation counts.

This structured approach facilitates automated data extraction and analysis.

Usage Limits

SerpApi enforces request quotas to ensure stable performance:

- *Free Plan*: 250 searches per month.
- *Paid Plans*: Higher monthly limits (5,000, 10,000, or more).
- *Rate Limits*: Requests per hour are capped to prevent abuse.

Usage limits balance accessibility with system reliability and provide flexible scaling for academic or enterprise needs.

Code Example.

Java:

```
import java.io.*;    // For InputStream, InputStreamReader, BufferedReader
import java.net.*;    // For URL and HttpURLConnection

public class ScholarSearchExample {
    public static void main(String[] args) throws Exception {
        // Your personal API key from SerpApi (replace "YOUR_API_KEY" with the real
        one)
        String apiKey = "YOUR_API_KEY";

        // The search query you want to run in Google Scholar
        // Spaces are replaced with '+' to make it URL-safe
        String query = "deep+learning";

        // Full request URL to SerpApi, specifying:
        // - engine=google_scholar (use Google Scholar engine)
        // - q=deep+learning (the query)
        // - hl=en (set language to English)
        // - api_key=YOUR_API_KEY (authentication)
        String urlStr = "https://serpapi.com/search?engine=google_scholar&q="
            + query + "&hl=en&api_key=" + apiKey;

        // Create a URL object from the string
        URL url = new URL(urlStr);

        // Open a connection to the URL
        HttpURLConnection conn = (HttpURLConnection) url.openConnection();

        // Specify that this is a GET request (retrieving information, not sending data)
        conn.setRequestMethod("GET");
```

```

        // Create a BufferedReader to read the response from the input stream
        BufferedReader in = new BufferedReader(new
InputStreamReader(conn.getInputStream()));

        // Read the response line by line
        String inputLine;
        StringBuilder content = new StringBuilder();
        while ((inputLine = in.readLine()) != null) {
            content.append(inputLine); // Append each line to the result string
        }

        // Close the reader and the HTTP connection
        in.close();
        conn.disconnect();

        // Print out the raw JSON response returned by the API
        System.out.println("Response JSON: " + content.toString());
    }
}

```

GITHUB:

https://github.com/DanaPonceDA/Server-and-Database-Commands_PonceDelAngel

SerpApi

Choose a Plan



Your email address has been successfully confirmed.



Free Plan



250 searches / month



Non-commercial use only

For free plans, we need to verify your account:



You have verified your email



You need to verify your phone number -

[Verify phone number](#)

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